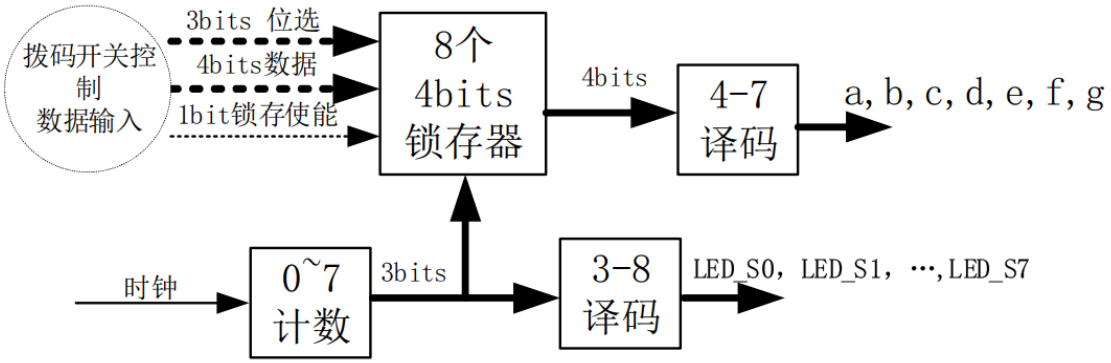
实验2实验报告

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1.实验设计



如图为七段数码管动态扫描控制电路的设计思路。需要设计计数器，4-7译码器，3-8译码器，分频器模块。其中将锁存器，计数器与3-8

译码器封装到一个模块。

2.代码

```
`timescale 10ns/1ns

module FPGA_EXP2_ly(
    input rst2,
    input clk,
    input en,
    input [3:0] input_data,
    input [2:0] reg_choose,
    output DP,
    output [6:0] LED_data,//reg[6]->a,reg[0]->g
    output reg[7:0] LED_choose,
    output clkout_debug
);

reg [3:0] data_S0;
reg [3:0] data_S1;
reg [3:0] data_S2;
reg [3:0] data_S3;
reg [3:0] data_S4;
reg [3:0] data_S5;
reg [3:0] data_S6;
reg [3:0] data_S7;

reg [3:0] output_data;

wire rst;
assign rst=~rst2;
assign DP=1'b0;
wire clkout;
```

```
reg [2:0] num;

div50MHZ div1(
    .clk(clk),
    .clkout(clkout),
    .rst(rst)
);

assign clkout_debug=clkout;
always @(posedge en) begin

    if(rst)
        begin
            data_S0<=4'b0001;
            data_S1<=4'b0010;
            data_S2<=4'b0011;
            data_S3<=4'b0100;
            data_S4<=4'b0101;
            data_S5<=4'b0110;
            data_S6<=4'b0111;
            data_S7<=4'b1000;
        end

    if(en)
        begin
            case (reg_choose)
                3'b000: data_S0<=input_data;
                3'b001: data_S1<=input_data;
                3'b010: data_S2<=input_data;
                3'b011: data_S3<=input_data;
                3'b100: data_S4<=input_data;
                3'b101: data_S5<=input_data;
                3'b110: data_S6<=input_data;
                3'b111: data_S7<=input_data;
            endcase
        end
    end

    always @(posedge clkout)
    begin
        if(num == 3'b111)
            begin
                num <= 3'b000;
            end
        else if(num>=3'b000 && num<=3'b110)
            begin
                num <= num + 3'b001;
            end
        else
            begin
                num <=3'b000;
            end
            //3-8译码器
        case (num)
        3'b000: begin output_data<=data_S0; LED_choose<=8'b00000001; end
        3'b001: begin output_data<=data_S1; LED_choose<=8'b00000010; end
        3'b010: begin output_data<=data_S2; LED_choose<=8'b00000100; end
        3'b011: begin output_data<=data_S3; LED_choose<=8'b00001000; end
```

```
        3'b100: begin output_data<=data_S4; LED_choose<=8'b00010000; end
        3'b101: begin output_data<=data_S5; LED_choose<=8'b00100000; end
        3'b110: begin output_data<=data_S6; LED_choose<=8'b01000000; end
        3'b111: begin output_data<=data_S7; LED_choose<=8'b10000000; end
        endcase
    end

LED_decoder decoder1(
    .in(output_data),
    .out(LED_data)
);

endmodule
//4-7译码器
module LED_decoder(
    input [3:0] in,
    output reg [6:0] out
);
    always @(*) begin
        case (in)
            4'b0000: out = 7'b0000001;
            4'b0001: out = 7'b1001111;
            4'b0010: out = 7'b0010010;
            4'b0011: out = 7'b0000110;
            4'b0100: out = 7'b1001100;
            4'b0101: out = 7'b0100100;
            4'b0110: out = 7'b0100000;
            4'b0111: out = 7'b0001111;
            4'b1000: out = 7'b0000000;
            4'b1001: out = 7'b0000100;
            4'b1010: out = 7'b0001000;
            4'b1011: out = 7'b1100000;
            4'b1100: out = 7'b0110001;
            4'b1101: out = 7'b1000010;
            4'b1110: out = 7'b0110000;
            4'b1111: out = 7'b0111000;
        endcase
    end
endmodule
//分频器
module div50MHZ(
    input clk,
    input rst,
    output reg clkout
);
    integer cnt;
    always @(posedge clk or posedge rst) begin
        if(rst)
        begin
            cnt <= 0;
            clkout <= 1'b0;
        end
        if(clk)
        begin
            if(cnt <= 24999 && cnt>= 0)
                begin
                    clkout <= 1'b1;
                    cnt <= cnt + 1;
```

```
                end
            else if(cnt >=25000 && cnt <= 49999)
                begin
                    clkout <= 1'b0;
                    cnt <= cnt + 1;
                end
            else
                begin
                    cnt<=0;
                end
        end

    end
endmodule
```

3.实验结果

2.1 仿真波形

激励文件：

```
`timescale 10ns/1ns

module FPGA_EXP2_ly_tb();
reg rst2;
reg clk;
reg en;
reg [3:0] input_data;
reg [2:0] reg_choose;
wire [6:0] LED_data;//reg[6]->a,reg[0]->g
wire [7:0] LED_choose;
wire clkout_debug;

FPGA_EXP2_ly fpga(
    .rst2(rst2),
    .clk(clk),
    .en(en),
    .input_data(input_data),
    .reg_choose(reg_choose),
    .LED_data(LED_data),
    .LED_choose(LED_choose),
    .clkout_debug(clkout_debug)
);

initial begin
    rst2=1;en=0;
    rst2=0;
    #5 rst2=1;
    #5 en=1;
    input_data=4'b0001;
    reg_choose=3'b000;
    #5 en=0;
    #5 en=1;
    input_data=4'b0010;
    reg_choose=3'b001;
    #5 en=0;
    #5 en=1;
```

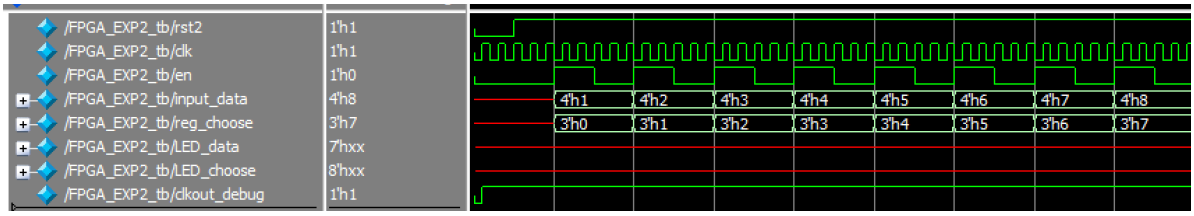
```
    input_data=4'b0011;
    reg_choose=3'b010;
    #5 en=0;
    #5 en=1;
    input_data=4'b0100;
    reg_choose=3'b011;
    #5 en=0;
    #5 en=1;
    input_data=4'b0101;
    reg_choose=3'b100;
    #5 en=0;
    #5 en=1;
    input_data=4'b0110;
    reg_choose=3'b101;
    #5 en=0;
    #5 en=1;
    input_data=4'b0111;
    reg_choose=3'b110;
    #5 en=0;
    #5 en=1;
    input_data=4'b1000;
    reg_choose=3'b111;
    #5
    en=0;
end

initial clk = 1'b0;
always #1 clk = ~clk; //产生时钟信号

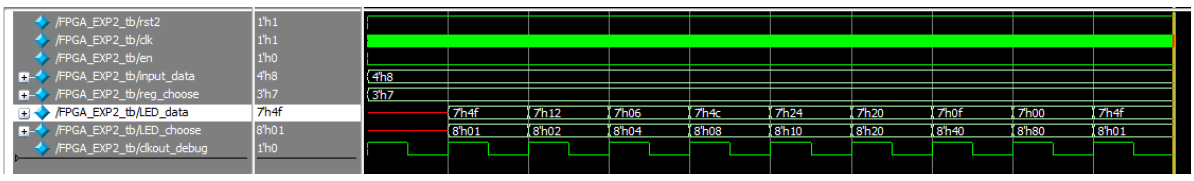
endmodule
```

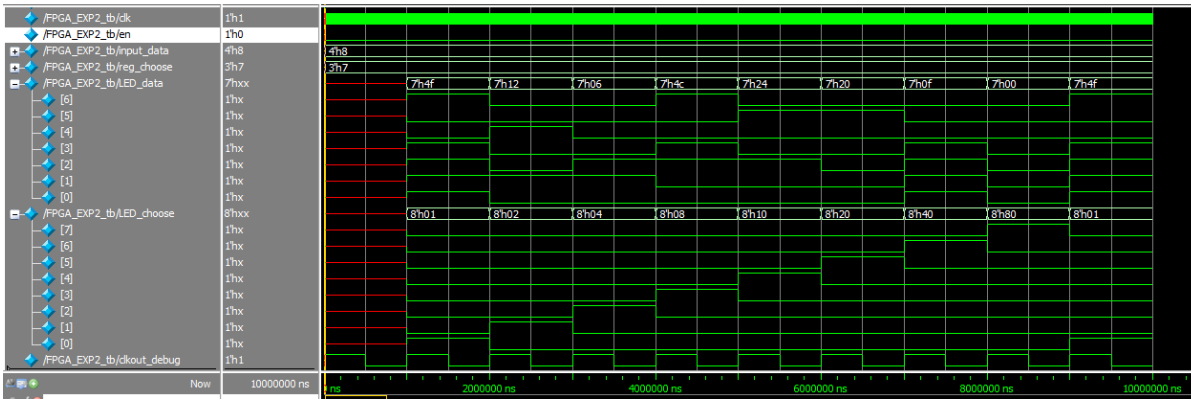
仿真波形

寄存器初值输入：



数码管动态扫描部分：





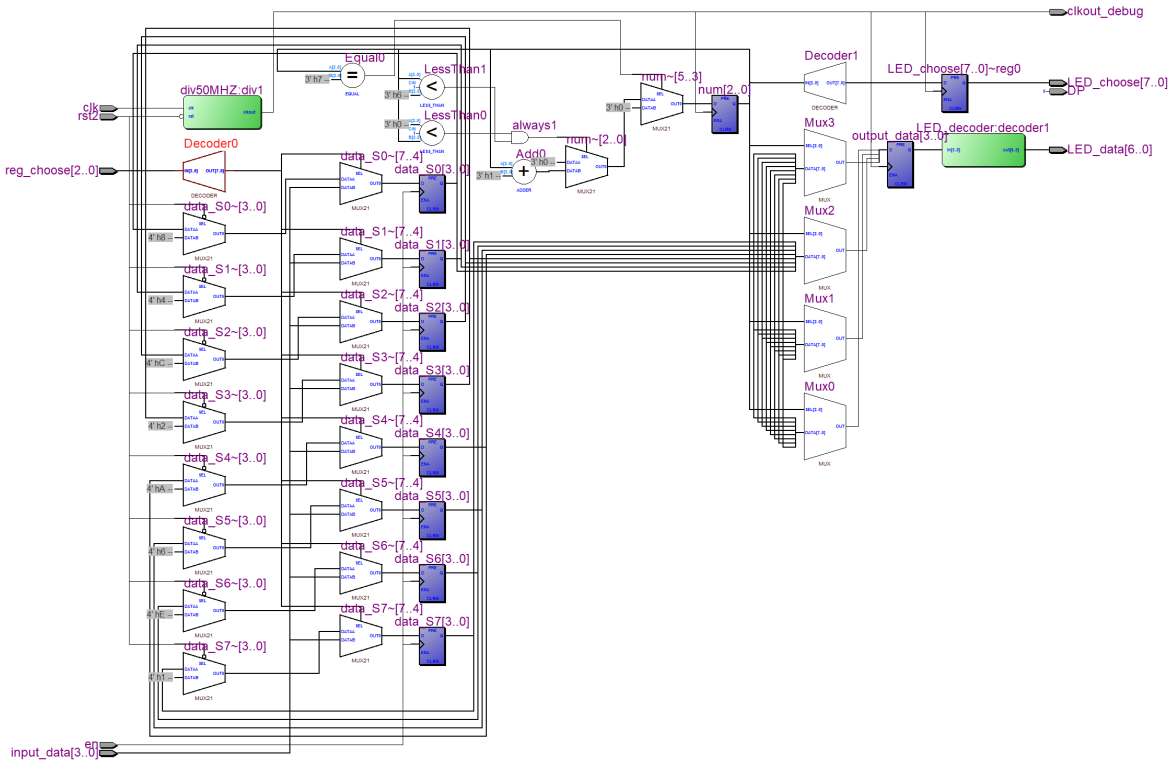
可以看到，数据被输入到了寄存器并正常输出

2.2 RTL电路

全编译之后的资源占用信息：

Flow Summary	
Flow Status	Successful - Mon Oct 21 21:59:44 2024
Quartus II 64-Bit Version	13.0.1 Build 232 06/12...3 SP 1 SJ Full Version
Revision Name	FPGA_EXP2
Top-level Entity Name	FPGA_EXP2
Family	Cyclone V
Device	5CEFA2F23C8
Timing Models	Final
Logic utilization (in ALMs)	96 / 9,430 (1 %)
Total registers	49
Total pins	27 / 224 (12 %)
Total virtual pins	0
Total block memory bits	0 / 1,802,240 (0 %)
Total DSP Blocks	0 / 25 (0 %)
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI PMA RX ATT Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total HSSI PMA TX ATT Serializers	0
Total PLLs	0 / 4 (0 %)
Total DLLs	0 / 4 (0 %)

RTL电路：

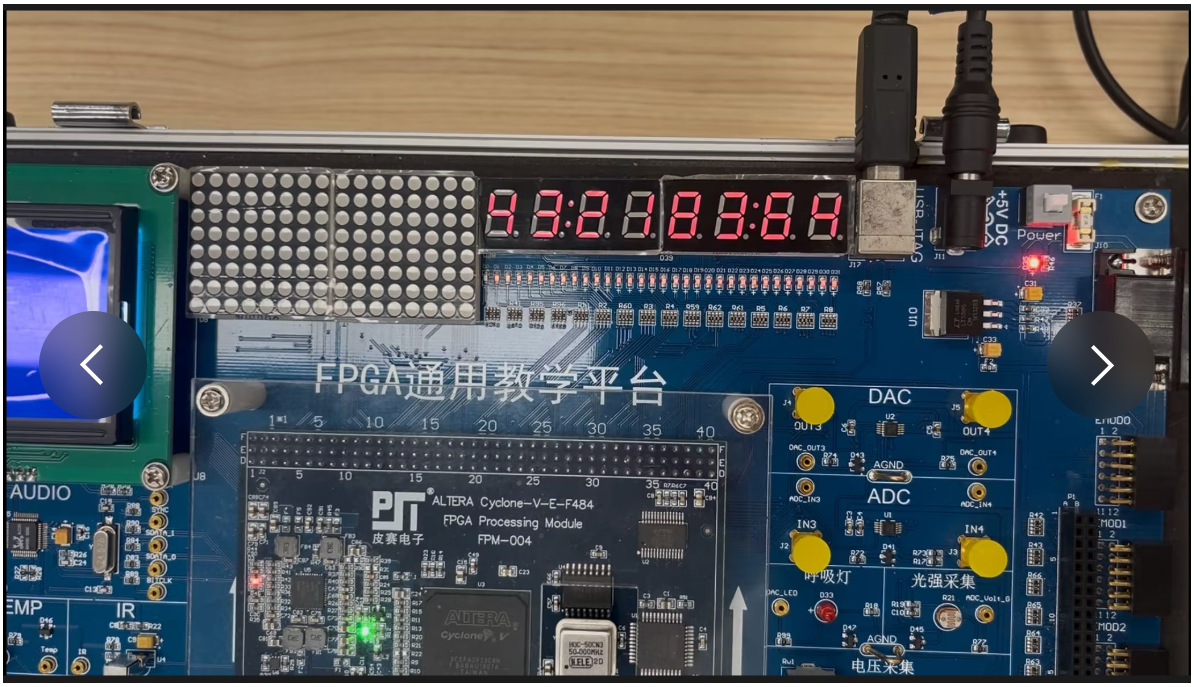


2.3 上板结果

管脚分配：

Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	Reserved	I/O Standard	Current Strength	Slew Rate
DP	Output	PIN_H16	7A	B7A_N0	PIN_H16		2.5 V (default)	12mA (default)	1 (default)
LED_choose[0]	Output	PIN_C11	7A	B7A_N0	PIN_C11		2.5 V (default)	12mA (default)	1 (default)
LED_choose[1]	Output	PIN_A13	7A	B7A_N0	PIN_A13		2.5 V (default)	12mA (default)	1 (default)
LED_choose[2]	Output	PIN_D13	7A	B7A_N0	PIN_D13		2.5 V (default)	12mA (default)	1 (default)
LED_choose[3]	Output	PIN_E12	7A	B7A_N0	PIN_E12		2.5 V (default)	12mA (default)	1 (default)
LED_choose[4]	Output	PIN_B16	7A	B7A_N0	PIN_B16		2.5 V (default)	12mA (default)	1 (default)
LED_choose[5]	Output	PIN_B15	7A	B7A_N0	PIN_B15		2.5 V (default)	12mA (default)	1 (default)
LED_choose[6]	Output	PIN_E14	7A	B7A_N0	PIN_E14		2.5 V (default)	12mA (default)	1 (default)
LED_choose[7]	Output	PIN_F13	7A	B7A_N0	PIN_F13		2.5 V (default)	12mA (default)	1 (default)
LED_data[0]	Output	PIN_G17	7A	B7A_N0	PIN_G17		2.5 V (default)	12mA (default)	1 (default)
LED_data[1]	Output	PIN_G13	7A	B7A_N0	PIN_G13		2.5 V (default)	12mA (default)	1 (default)
LED_data[2]	Output	PIN_E16	7A	B7A_N0	PIN_E16		2.5 V (default)	12mA (default)	1 (default)
LED_data[3]	Output	PIN_M18	5B	B5B_N0	PIN_M18		2.5 V (default)	12mA (default)	1 (default)
LED_data[4]	Output	PIN_K20	7A	B7A_N0	PIN_K20		2.5 V (default)	12mA (default)	1 (default)
LED_data[5]	Output	PIN_H18	7A	B7A_N0	PIN_H18		2.5 V (default)	12mA (default)	1 (default)
LED_data[6]	Output	PIN_K19	7A	B7A_N0	PIN_K19		2.5 V (default)	12mA (default)	1 (default)
clk	Input	PIN_W16	4A	B4A_N0	PIN_W16		2.5 V (default)	12mA (default)	
clkout_debug	Output	PIN_D12	7A	B7A_N0	PIN_D12		2.5 V (default)	12mA (default)	1 (default)
en	Input	PIN_Y11	3B	B3B_N0	PIN_Y11		2.5 V (default)	12mA (default)	
input_data[0]	Input	PIN_AB13	4A	B4A_N0	PIN_AB13		2.5 V (default)	12mA (default)	
input_data[1]	Input	PIN_V13	4A	B4A_N0	PIN_V13		2.5 V (default)	12mA (default)	
input_data[2]	Input	PIN_U12	3B	B3B_N0	PIN_U12		2.5 V (default)	12mA (default)	
input_data[3]	Input	PIN_R11	3B	B3B_N0	PIN_R11		2.5 V (default)	12mA (default)	
reg_choose[0]	Input	PIN_AA15	4A	B4A_N0	PIN_AA15		2.5 V (default)	12mA (default)	
reg_choose[1]	Input	PIN_Y15	4A	B4A_N0	PIN_Y15		2.5 V (default)	12mA (default)	
reg_choose[2]	Input	PIN_Y14	4A	B4A_N0	PIN_Y14		2.5 V (default)	12mA (default)	
rst2	Input	PIN_N16	5B	B5B_N0	PIN_N16		2.5 V (default)	12mA (default)	
<<new node>>									

上板结果：



如图所示，可以正常刷新显示。

4.实验总结

本次实验完整体验熟悉了FPGA开发流程，实验中发现上板数码管闪烁情况，经过排查是always语句内同时判断多个触发条件导致。经过

debug后运行正常。